

Subject: Azure Fundamentals

Unit 3 : Security, Privacy, Compliance, and Trust:

Computer Science & Engineering

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Outline

- Introduce the Azure security features
- Identity and access management
- **Azure Active Directory and data Protection**
- **Compliance frameworks**
- Azure governance methodologies

Introduce the Azure security features



Azure Active Directory (Azure AD)

Overview of Azure AD:

Azure AD Architecture Overview

- **Description:** Azure AD as the central hub connecting users, devices, applications (both cloud and on-premises), and on-premises AD through Azure AD Connect.
- **Components:**
 - **Users & Devices:** Represented by icons on the left side.
 - **Azure AD:** In the center, showing connections to different applications (Office 365, SaaS apps, etc.) and on-premises AD.
 - **Cloud Applications & On-Premises Resources:** Displayed on the right, connected to Azure AD.

Azure Active Directory (Azure AD)

Setting Up Azure AD:

Creating a New Azure AD Tenant

• **Description:** A step-by-step guide.

• **Steps:**

- Navigate to the Azure portal.
- Go to “Azure Active Directory.”
- Click “Create a tenant.”
- Enter the required details (Organization name, Initial domain name, Country/Region).
- Review and create.

Azure Active Directory (Azure AD)

Adding Users and Groups

•**Description:** how to add users manually or sync with on-premises AD using Azure AD Connect.

•**Steps:**

- Navigate to “Users” in Azure AD.
- Click “New User” and fill in the details.
- For syncing with on-premises AD, show how to set up Azure AD Connect.

Configuring SSO

•**Description:** Screenshots or diagrams showing the configuration of SSO for a popular app like Office 365.

•**Steps:**

- Select an application from the “Enterprise applications” in Azure AD.
- Configure SSO settings (SAML, OAuth, etc.).
- Assign users to the application.

Azure Active Directory (Azure AD)

Enabling MFA

•**Description:** enabling MFA for users.

•**Steps:**

- Go to “Azure AD,” then “Security.”
- Select “Multi-Factor Authentication.”
- Enable MFA for users or groups.

3. Managing Access and Security:

Conditional Access Policies

•**Description:** A screenshot walkthrough of setting up a conditional access policy.

•**Steps:**

- Navigate to “Security” in Azure AD.
- Select “Conditional Access.”
- Create a new policy and configure conditions (e.g., location, device compliance).
- Define access controls (e.g., require MFA, block access).

Azure Active Directory (Azure AD)

Role-Based Access Control (RBAC)

• **Description:** how roles are assigned to users/groups in Azure AD and how they control access to resources.

• **Components:**

- Users/Groups on the left.
- Roles (e.g., Reader, Contributor, Owner) in the center.
- Resources (e.g., Virtual Machines, Storage Accounts) on the right.

The image features a close-up, high-angle view of a brown printed circuit board (PCB) with intricate white circuit traces. A silver-colored metal padlock is positioned diagonally across the center-left of the frame. The background is filled with a dense pattern of white binary code (0s and 1s) overlaid on the circuit traces, creating a digital and technological aesthetic. The lighting is soft, highlighting the metallic texture of the padlock and the fine details of the PCB.

SECURITY, PRIVACY, COMPLIANCE AND TRUST

Data Protection Compliance Frameworks

Compliance Framework Mapping

- **Description:** A table or matrix mapping different compliance frameworks (e.g., GDPR, HIPAA, ISO/IEC 27001) to key Azure services (e.g., Azure Information Protection, Azure Policy, Azure Key Vault).
- **Components:**
 - Columns representing Compliance Frameworks.
 - Rows representing Azure Services.
 - Checkmarks indicating which services align with which frameworks.

Data Protection Compliance Frameworks

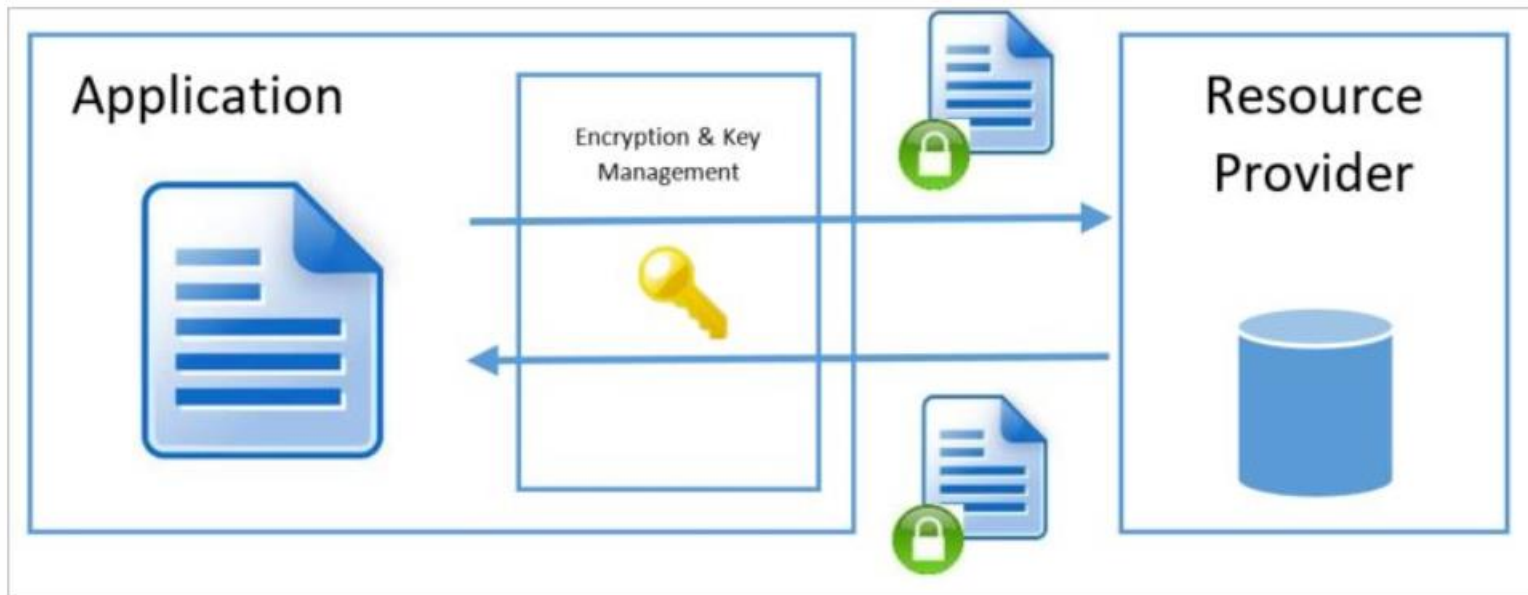
Implementing Data Protection in Azure:

Data Encryption in Azure

- **Description:** how data is encrypted at rest and in transit using Azure services.
- **Components:**
 - **Data at Rest:** Show data stored in Azure Storage or SQL Database, with encryption (e.g., Azure Key Vault, TDE) applied.
 - **Data in Transit:** Display data moving between services (e.g., via SSL/TLS) with encryption.

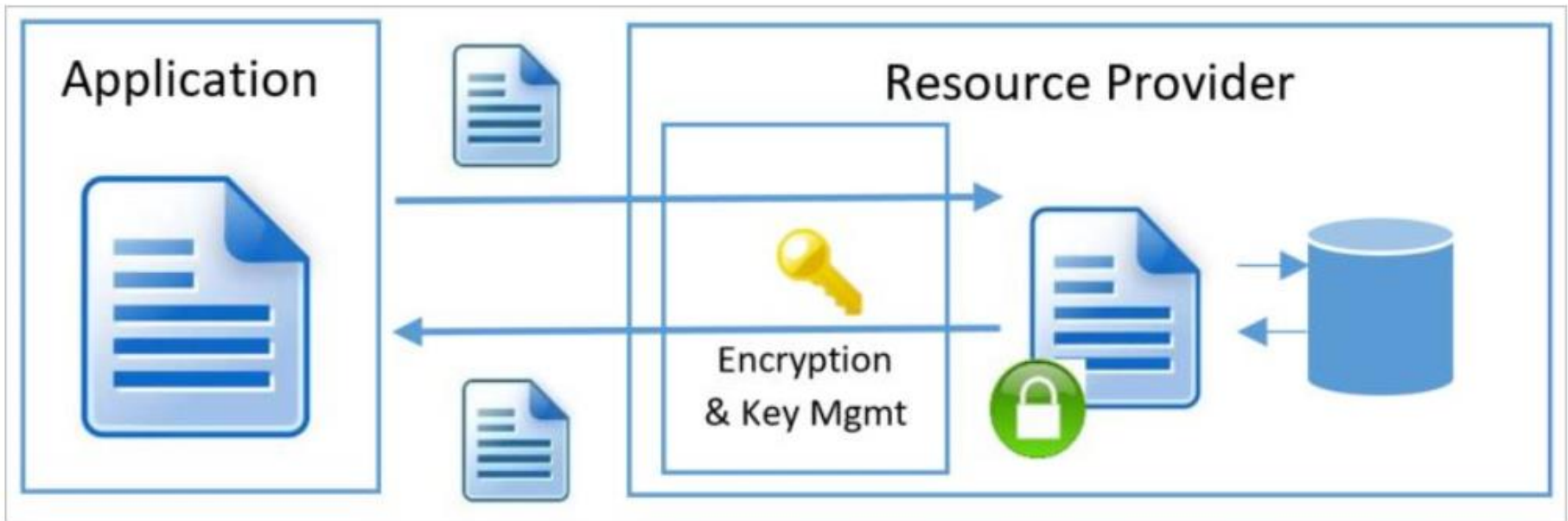
Data Protection Compliance Frameworks

Client-Side Encryption



Data Protection Compliance Frameworks

Server-Side Encryption

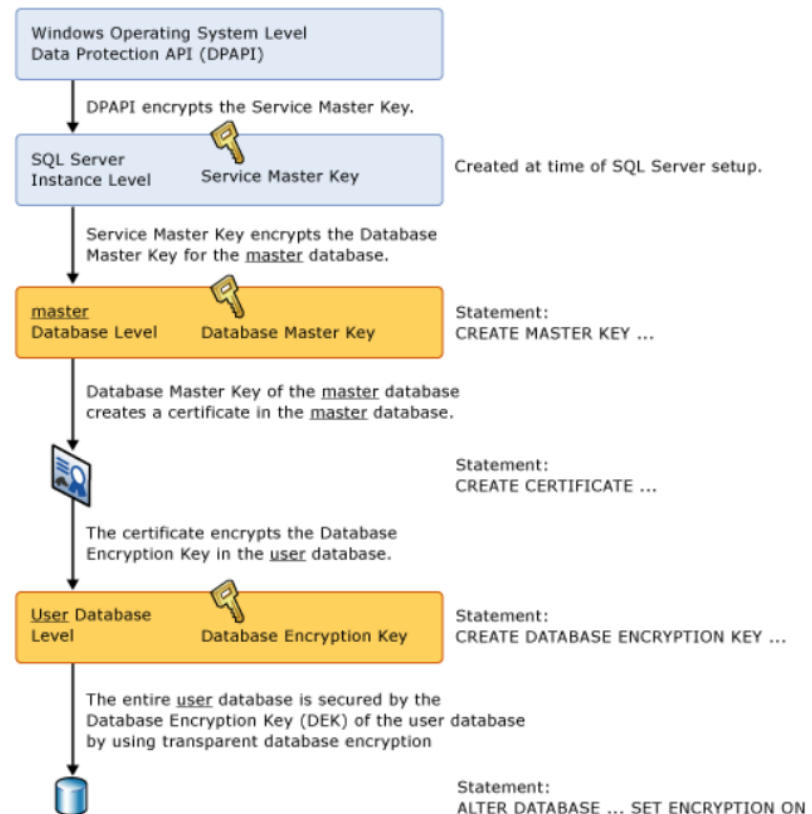


Data Protection Compliance Frameworks

Transparent Data Encryption (TDE)



Transparent Database Encryption Architecture



Data Protection Compliance Frameworks

Data Classification and Labeling

•Steps:

- Navigate to “Azure Information Protection.”
- Create and configure labels for different data sensitivity levels (e.g., Confidential, Internal).
- Apply labels to documents or emails.

3. Compliance Management in Azure: Using Azure Policy

Steps:

- Go to “Azure Policy” in the Azure portal.
- Create a new policy definition (e.g., enforce encryption on storage accounts).
- Assign the policy to a scope (e.g., a subscription or resource group).

Data Protection Compliance Frameworks



Data Protection Compliance Frameworks

Monitoring and Auditing with Azure Security Center

- **Description:** the dashboard in Azure Security Center for monitoring compliance and security posture.
- **Components:**
 - Compliance status summary.
 - Security alerts.
 - Recommendations.

Azure Governance Methodologie

- Introduce the Azure security features
- Identity and access management
- **Azure Active Directory and data Protection**
- **Compliance frameworks**
- Azure governance methodologies



Zero Trust and Defense in Depth

- Choosing a reliable cybersecurity solution that prevents security breaches and ensures secure remote access to employees is highly important.
- With the increasing cybersecurity concerns, organizations have started enforcing network security precautions, mainly because the organization's infrastructure and facilities are the common targets of cyber attackers.
- These cyberattacks and ransomware attacks lead to financial losses for organizations along with reputational damage. Hence, employing the right security strategy to prevent these attacks and ensures the utmost network security is paramount.

Zero Trust and Defense in Depth

It is based on the concept of “never trust, always verify.” This cybersecurity framework is based on the idea that no entity – inside or outside of a network – should be implicitly trusted.

Everything and everyone needs to prove who they are and that they have the right permissions to access the target resource. In short, they need to undergo continual authentication, verification and authorization to keep access to applications and data.

This is akin to crew members of the Enterprise verifying and authenticating their identity at every doorway before they are allowed to access the next area.

Zero Trust

Zero trust is a network security model that has more specific principles than those of defense in depth. With zero trust, the goal is to never trust anyone or anything that accesses and communicates with applications and services inside a corporate network.

Zero trust uses the principle of least privilege to ensure that only the correct devices and users connecting to the network, as well as workloads within public and private data centers, are allowed and expected to transmit and receive data.

Steps to build a zero-trust network

Network teams are largely responsible for deploying and configuring the elements that make up a zero-trust network. But security teams should also be involved in developing the overall zero-trust architecture.



STEP 1

Identify users and devices that attempt to connect to the network

Tool used:

Identity and access management



STEP 2

Set up access controls for application, file and service access

Tool used:

Next-generation firewall to create microsegmentation



STEP 3

Deploy tools to continuously monitor the network and device behavior

Tool used:

Network detection and response, AI for IT operations



STEP 4

Evaluate remote access to ensure proper security and authentication

Tool used:

Remote access VPNs

Benefits of zero trust

- The benefit of zero trust is contained within the adage of "never trust, verify everything." With zero trust, users must constantly prove they have appropriate permissions, from initial login and thereafter.
- This means that, even if attackers infiltrate a system, they can't necessarily do any damage or exfiltrate data.
- Despite the restrictive-sounding nature of zero trust, it can boost UX through tools such as single sign-on, reducing the worry around password mismanagement.

- **UX(User Experience Design)** - A person's perceptions and responses that result from the use or anticipated use of a product, system or service
- **UI (User interface (UI) design)** is the process designers use to build interfaces in software or computerized devices, focusing on looks or style.

What is Defense in Depth

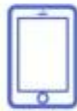
- Now, defense-in-depth refers to the practice of using layers of security measures, such as **firewalls, secured gateways, authentication and intrusion detection systems**, to protect internal networks from external attacks.
- This provides backup levels of security in case other security measures fail.
- With multiple layers of protective devices and processes to keep threat actors outside of the perimeter.
- In short, if one defensive measure is compromised, another set of defense mechanisms can detect and prevent a breach attempt.
- By including redundancies and using security defenses across solutions, enterprises can aim to close gaps in security and thwart potential attacks.

Introduce the Azure security features

7 Security features of Microsoft Cloud



Asset
Protection



Application
Protection



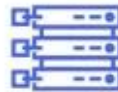
Access
Protection



Network
Security



Security Threat
Protection



Data
Encryption



Antivirus
Technology

Identity and access management

<https://learn.microsoft.com/en-us/azure/security/fundamentals/overview>

<https://www.microsoft.com/en-in/security/business/security-101/what-is-identity-access-management-iam#:~:text=Identity%20and%20access%20management%20is,who%20need%20access%20have%20access.>

<https://www.youtube.com/watch?v=7SjmCmEQE0I>

<https://www.youtube.com/watch?v=JI fvvekFwo>

Thank You!!!

x DIGITAL LEARNING CONTENT

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